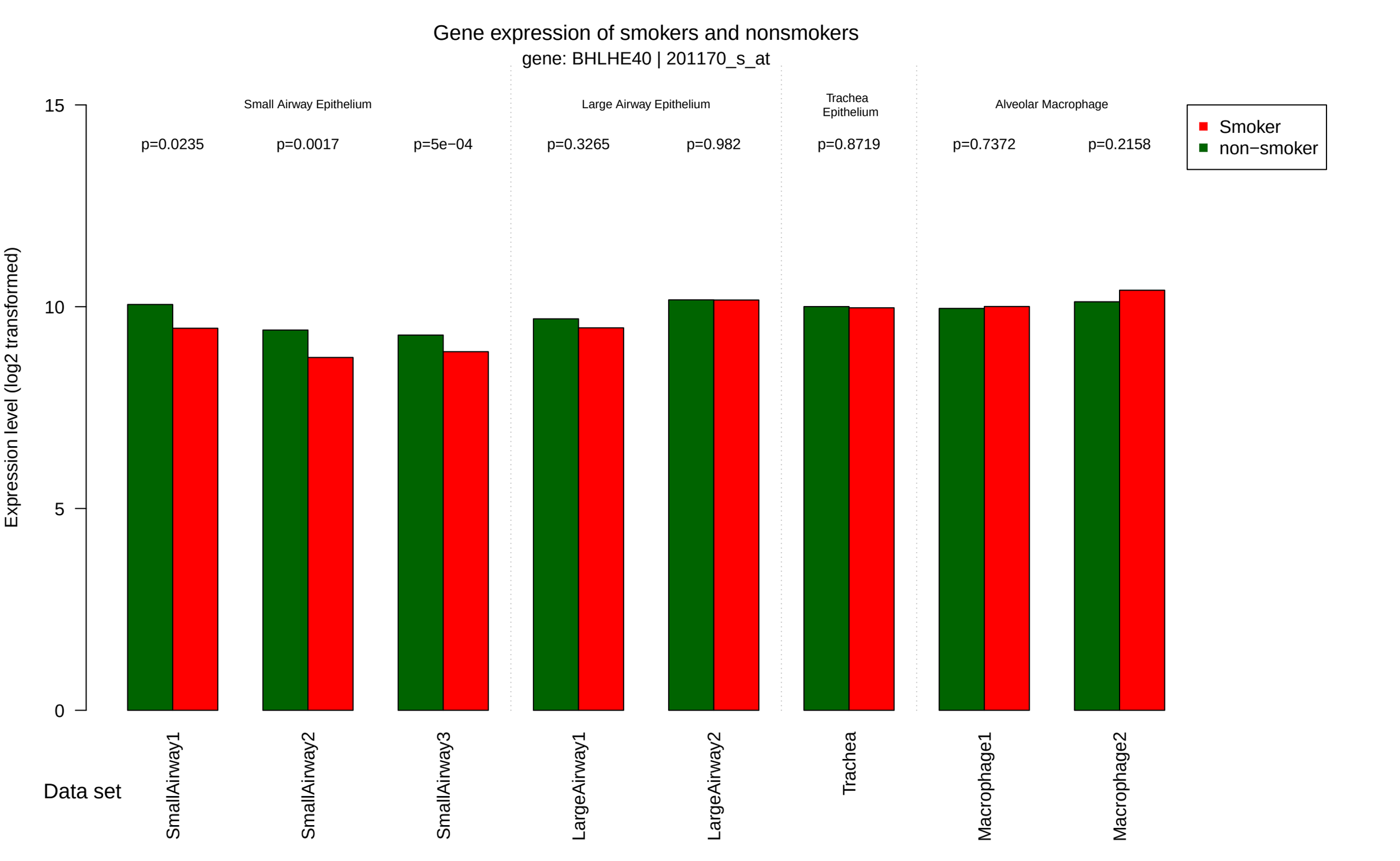
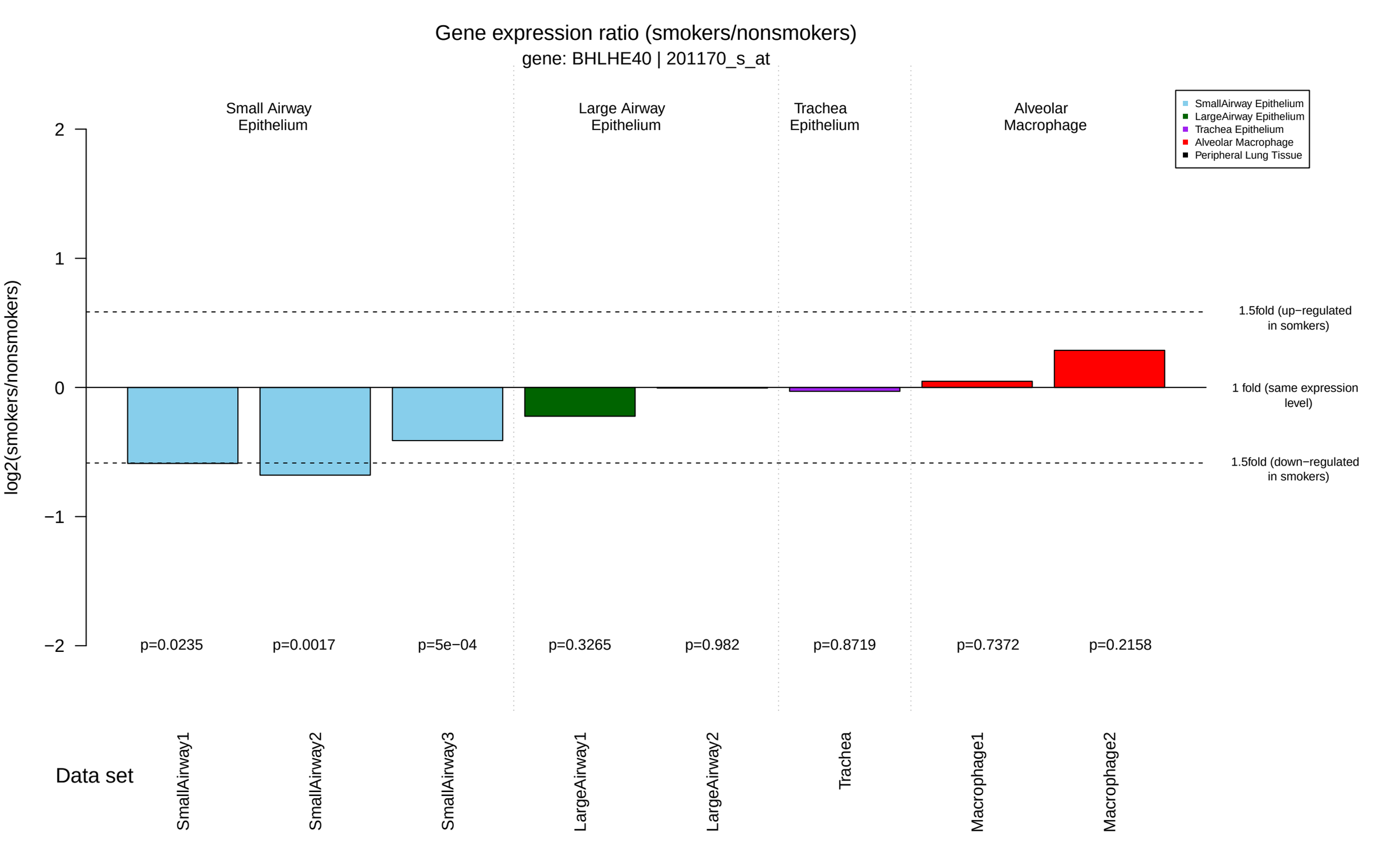
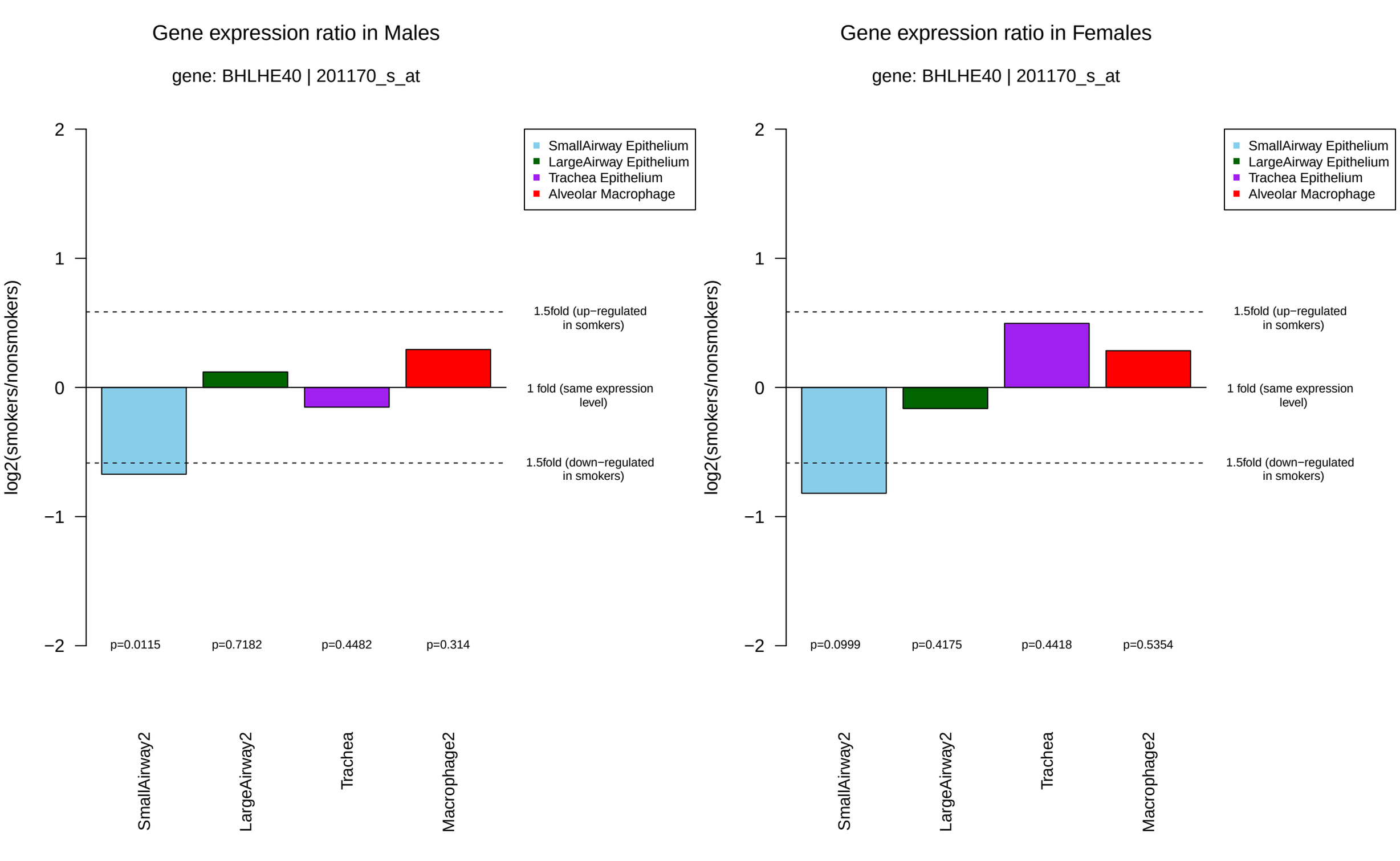
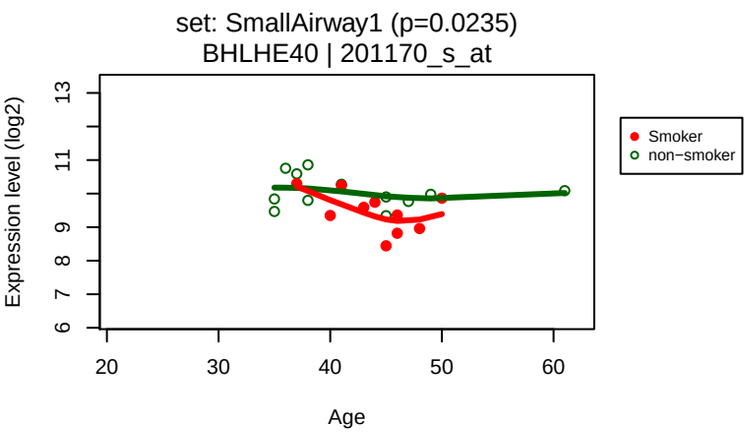
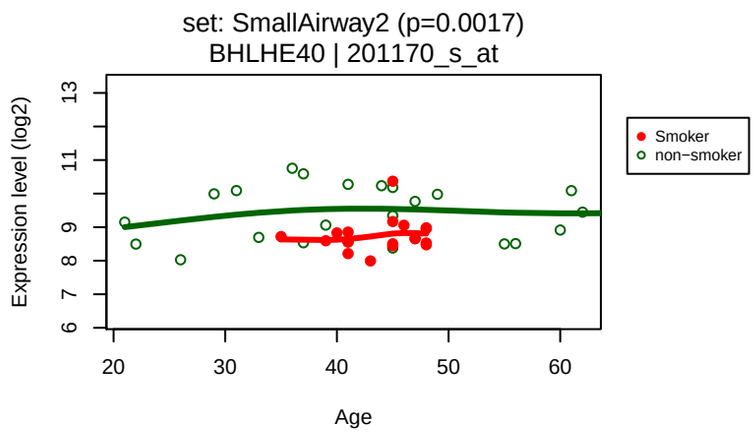
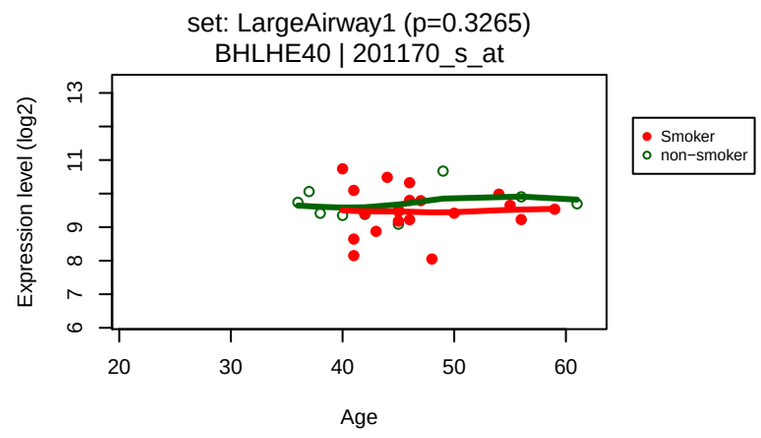

Gene expression of smokers and nonsmokers
gene: BHLHE40 | 201170_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0235
p=0.0017
p=5e−04
p=0.3265
p=0.982
p=0.8719
p=0.7372
p=0.2158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

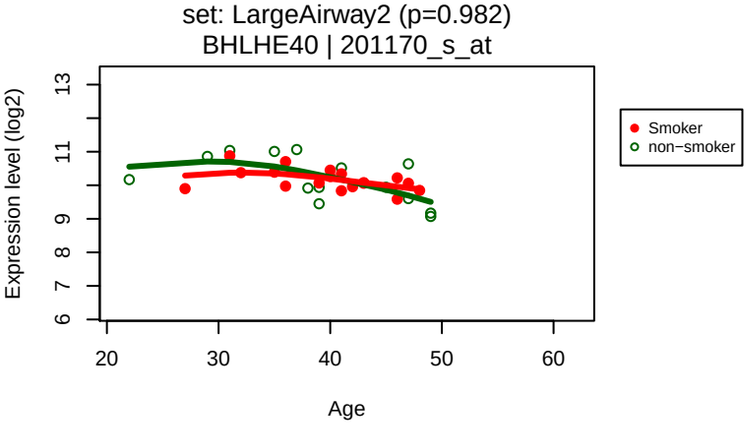

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE40 | 201170_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0235
p=0.0017
p=5e−04
p=0.3265
p=0.982
p=0.8719
p=0.7372
p=0.2158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

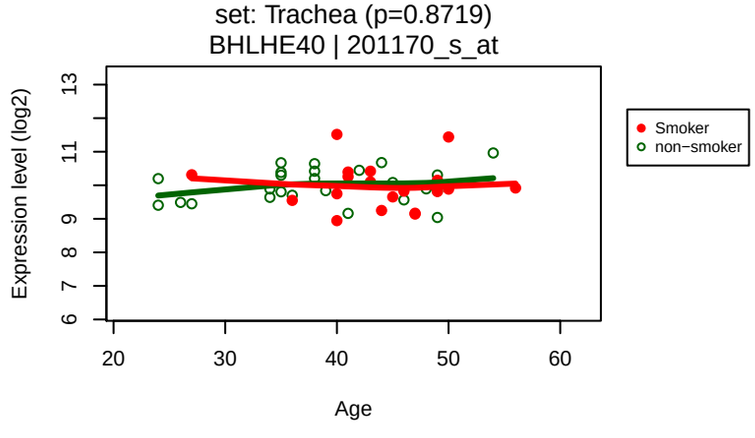

Gene expression ratio in Males
gene: BHLHE40 | 201170_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0115
p=0.7182
p=0.4482
p=0.314
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE40 | 201170_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0999
p=0.4175
p=0.4418
p=0.5354
SmallAirway2
LargeAirway2
Trachea
Macrophage2

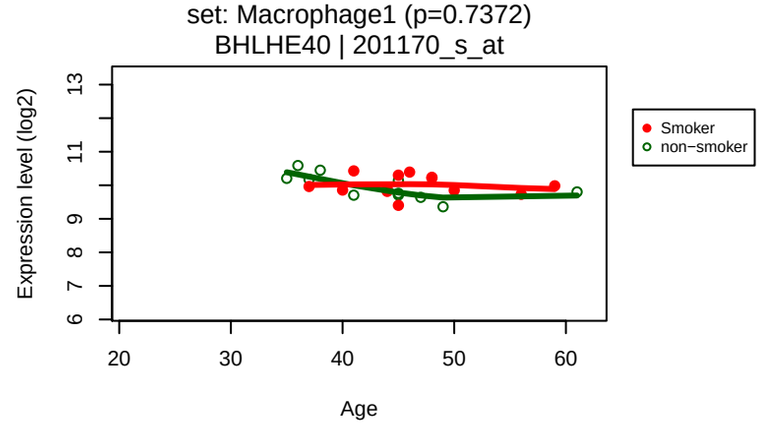

set: SmallAirway1 (p=0.0235)
BHLHE40 | 201170_s_at
Expression level (log2)
Age
Smoker
non−smoker

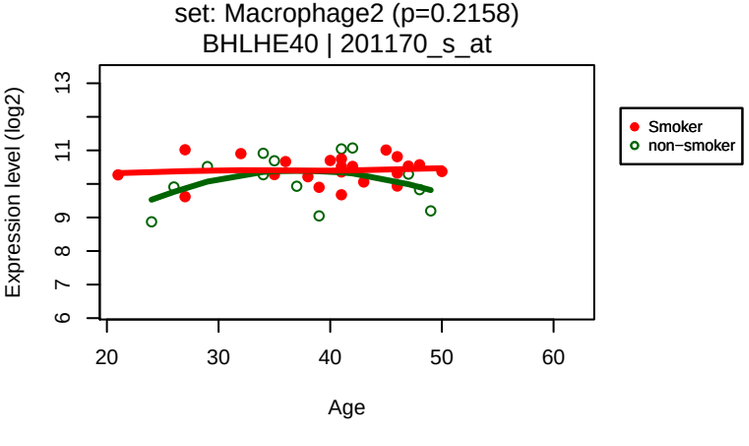

set: SmallAirway2 (p=0.0017)
BHLHE40 | 201170_s_at
Expression level (log2)
Age
Smoker
non−smoker

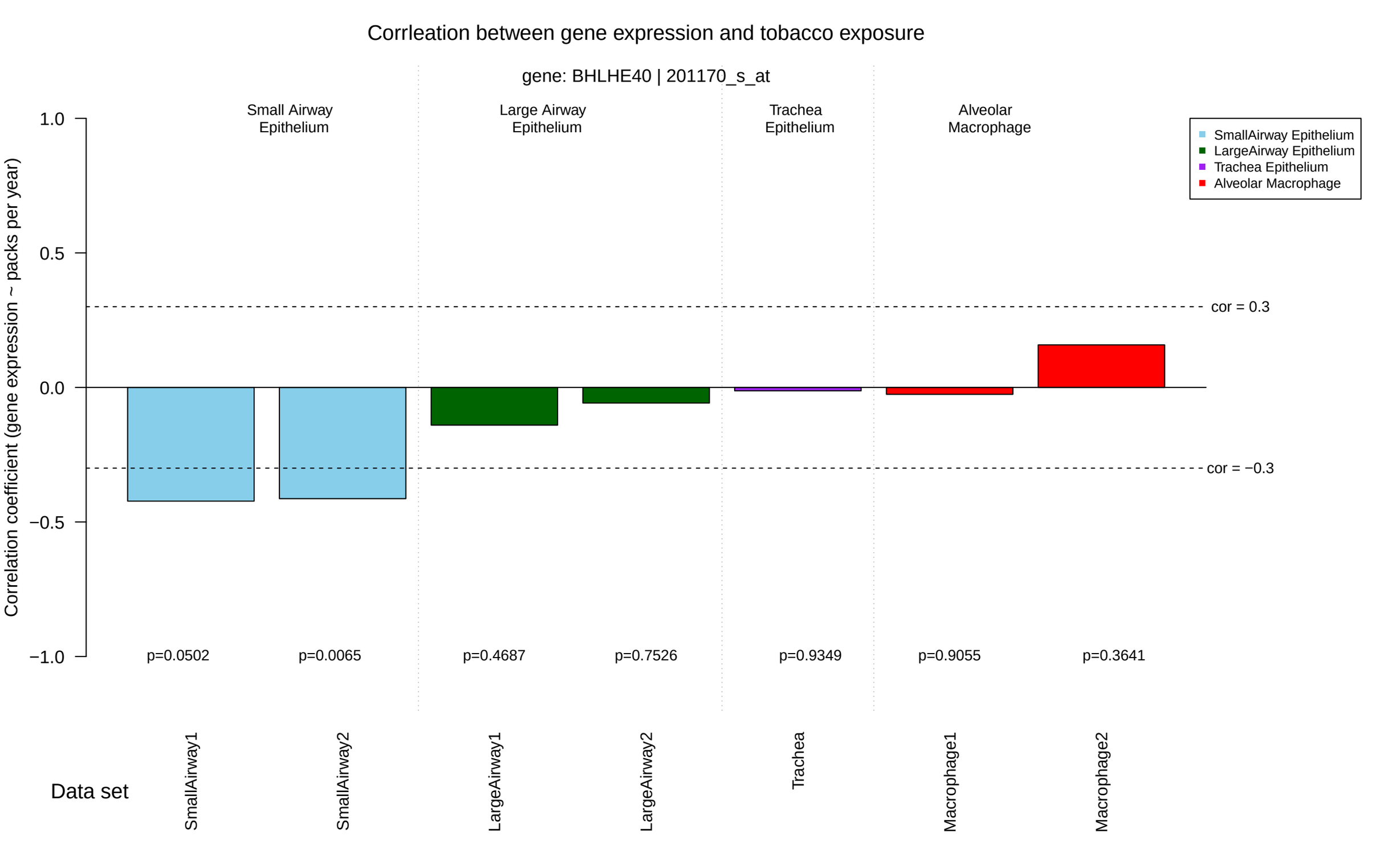

set: LargeAirway1 (p=0.3265)
BHLHE40 | 201170_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.982)
BHLHE40 | 201170_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8719)
BHLHE40 | 201170_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7372)
BHLHE40 | 201170_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2158)
BHLHE40 | 201170_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE40 | 201170_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0502
p=0.0065
p=0.4687
p=0.7526
p=0.9349
p=0.9055
p=0.3641
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2